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CMPE 256 Recommender System Data Science Hackathon

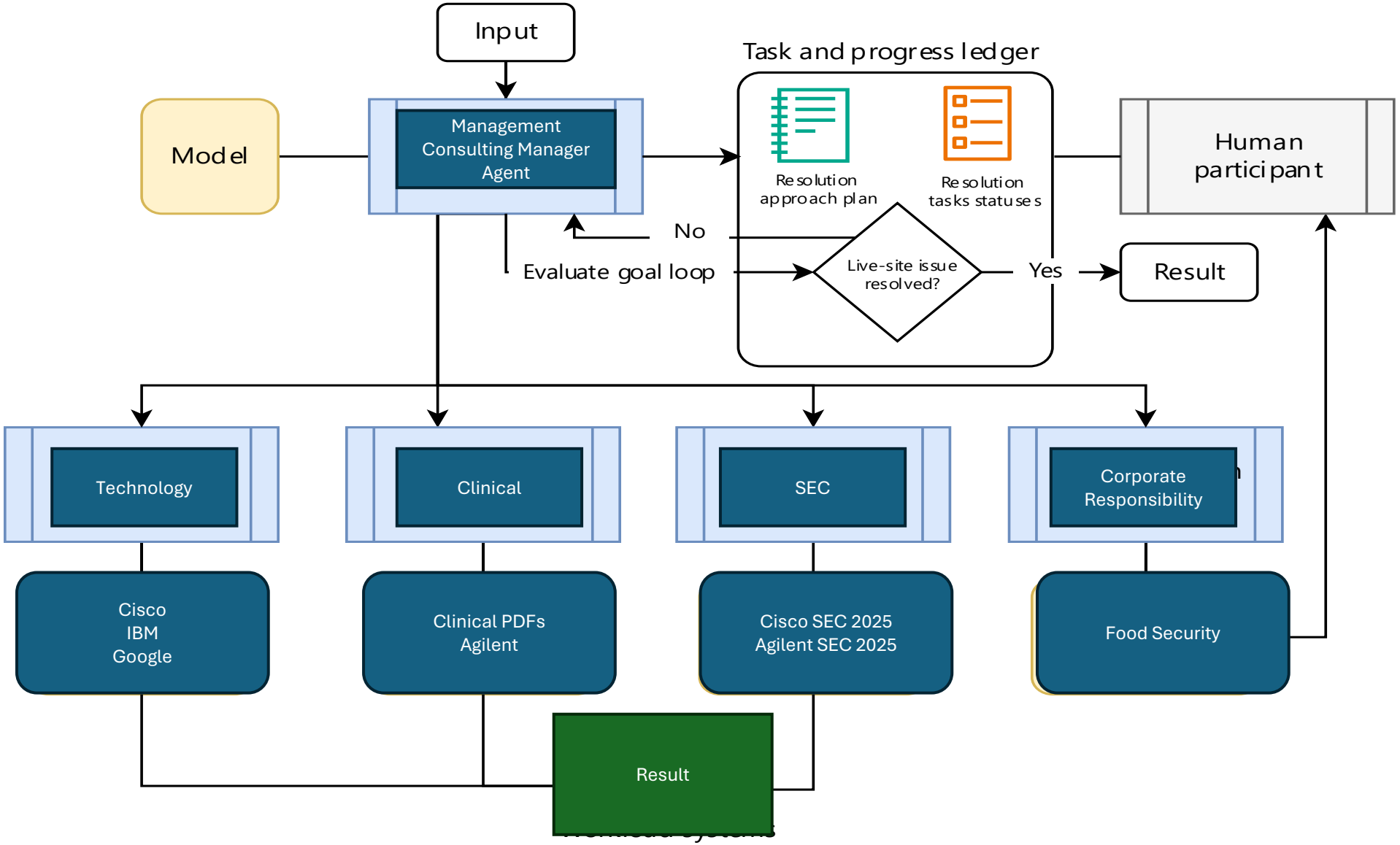
Dates: October 17,2026

Fall 2026

Professor: Chandrasekar Vuppalapati

Data Science Hackathon: Building an Agentic AI Management Consulting Platform

Multi Agent Architecture



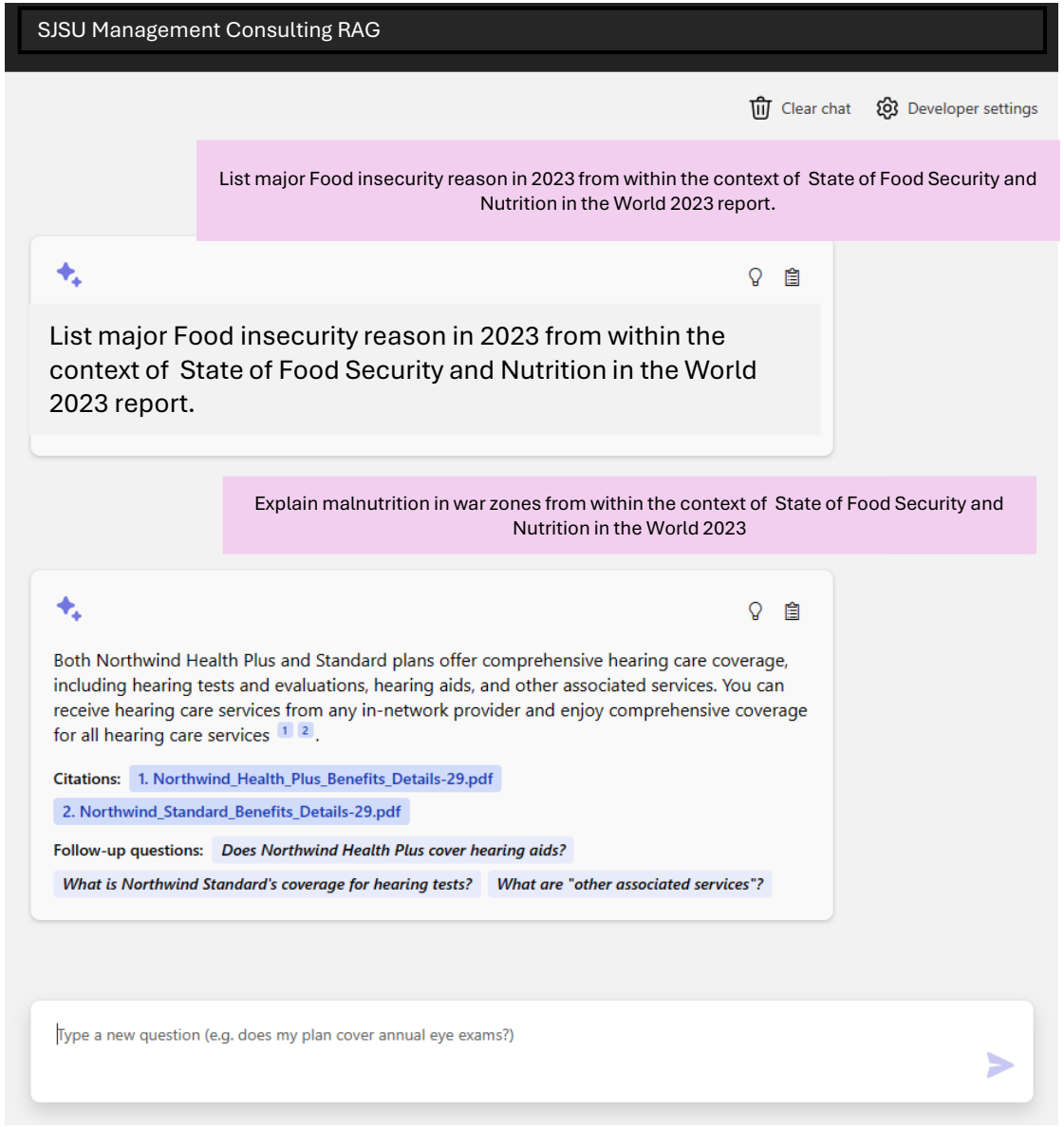
https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns

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Multi Agent Architecture

Prompts

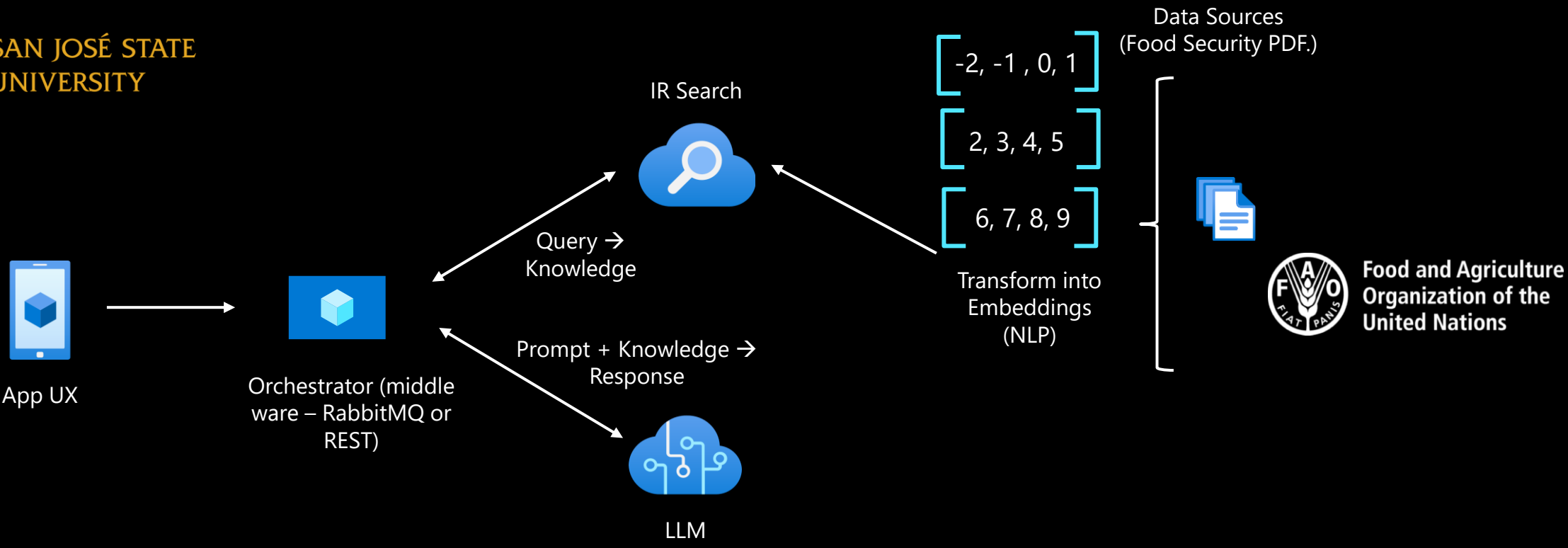
- List major Food insecurity reason in 2024
- Explain malnutrition in war zones
- Explain increase prices impact on food security



Enterprise Software Platform: Retrieval Augmented Generation

Food and Agriculture Organization of the United Nations – The State of Food Security and Nutrition in the World

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Market Basket Recommendation

Electronic Retailer Market Basket Recommendation System for Online Checkout Integration

As a Data Science student at San José State University, you have been approached by an electronic retailer to design and implement a Market Basket Recommendation System for their online checkout process. The primary objective is to enhance the customer's shopping experience by intelligently recommending additional products that are frequently purchased together.

When a shopper visits the retailer's website, the checkout cart begins empty. As soon as the customer adds the first item, the system should analyze historical transaction data to identify items that have been commonly purchased alongside that product. The model then generates a ranked list of recommendations, presenting them as helpful suggestions to the user.

Each time the customer adds a new item—either from the recommended list or by freely selecting a different product—the system dynamically updates its recommendations based on the current combination of items in the cart. This iterative recommendation process continues throughout the checkout experience, allowing the model to refine its predictions as new items are added.

Electronic Retailer Market Basket Recommendation System for Online Checkout Integration

In some instances, the combination of selected items may not exist in the historical transactions dataset. In such cases, the model should gracefully handle the absence of patterns and refrain from displaying irrelevant recommendations. This ensures that the system remains both accurate and contextually relevant.

As a Data Science major, your role involves designing, developing, and deploying this recommendation model. Once trained, the model can be deployed as a FAST API or microservice, integrated with the retailer's checkout system to deliver real-time recommendations. The deployment phase would also involve model validation, performance monitoring, and periodic retraining to ensure continued relevance as shopping trends evolve.

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Electronic Retailer Market Basket Recommendation System for Online Checkout Integration

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Electronic Item Checkout Web Portal

Added Items to Cart:

New User Comes to Portal

Select Item from the List:

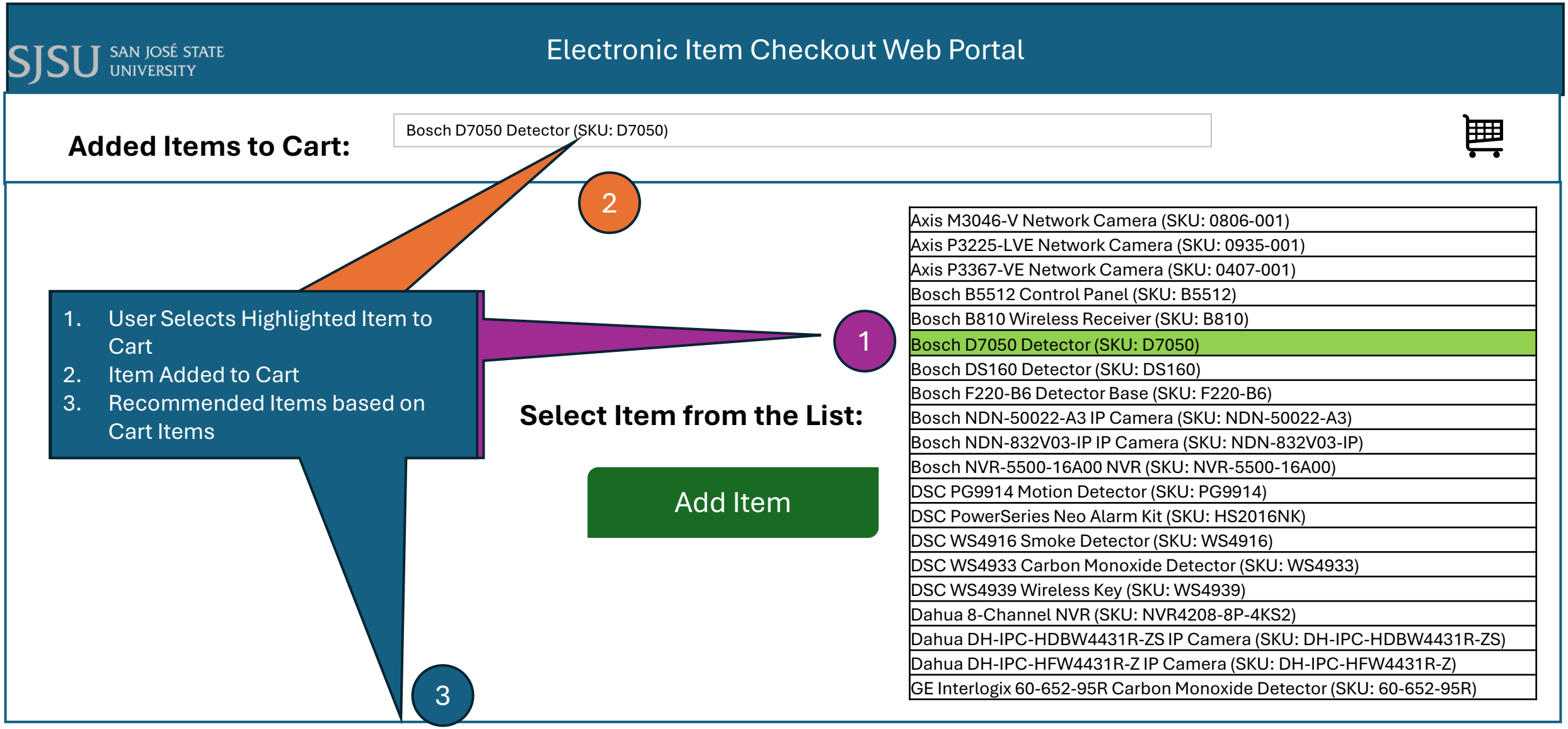
Add Item

Axis M3046-V Network Camera (SKU: 0806-001)
Axis P3225-LVE Network Camera (SKU: 0935-001)
Axis P3367-VE Network Camera (SKU: 0407-001)
Bosch B5512 Control Panel (SKU: B5512)
Bosch B810 Wireless Receiver (SKU: B810)
Bosch D7050 Detector (SKU: D7050)
Bosch DS160 Detector (SKU: DS160)
Bosch F220-B6 Detector Base (SKU: F220-B6)
Bosch NDN-50022-A3 IP Camera (SKU: NDN-50022-A3)
Bosch NDN-832V03-IP IP Camera (SKU: NDN-832V03-IP)
Bosch NVR-5500-16A00 NVR (SKU: NVR-5500-16A00)
DSC PG9914 Motion Detector (SKU: PG9914)
DSC PowerSeries Neo Alarm Kit (SKU: HS2016NK)
DSC WS4916 Smoke Detector (SKU: WS4916)
DSC WS4933 Carbon Monoxide Detector (SKU: WS4933)
DSC WS4939 Wireless Key (SKU: WS4939)
Dahua 8-Channel NVR (SKU: NVR4208-8P-4KS2)
Dahua DH-IPC-HDBW4431R-ZS IP Camera (SKU: DH-IPC-HDBW4431R-ZS)
Dahua DH-IPC-HFW4431R-Z IP Camera (SKU: DH-IPC-HFW4431R-Z)
GE Interlogix 60-652-95R Carbon Monoxide Detector (SKU: 60-652-95R)

Recommended Items:

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Electronic Retailer Market Basket Recommendation System for Online Checkout Integration



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Electronic Retailer Market Basket Recommendation System for Online Checkout Integration

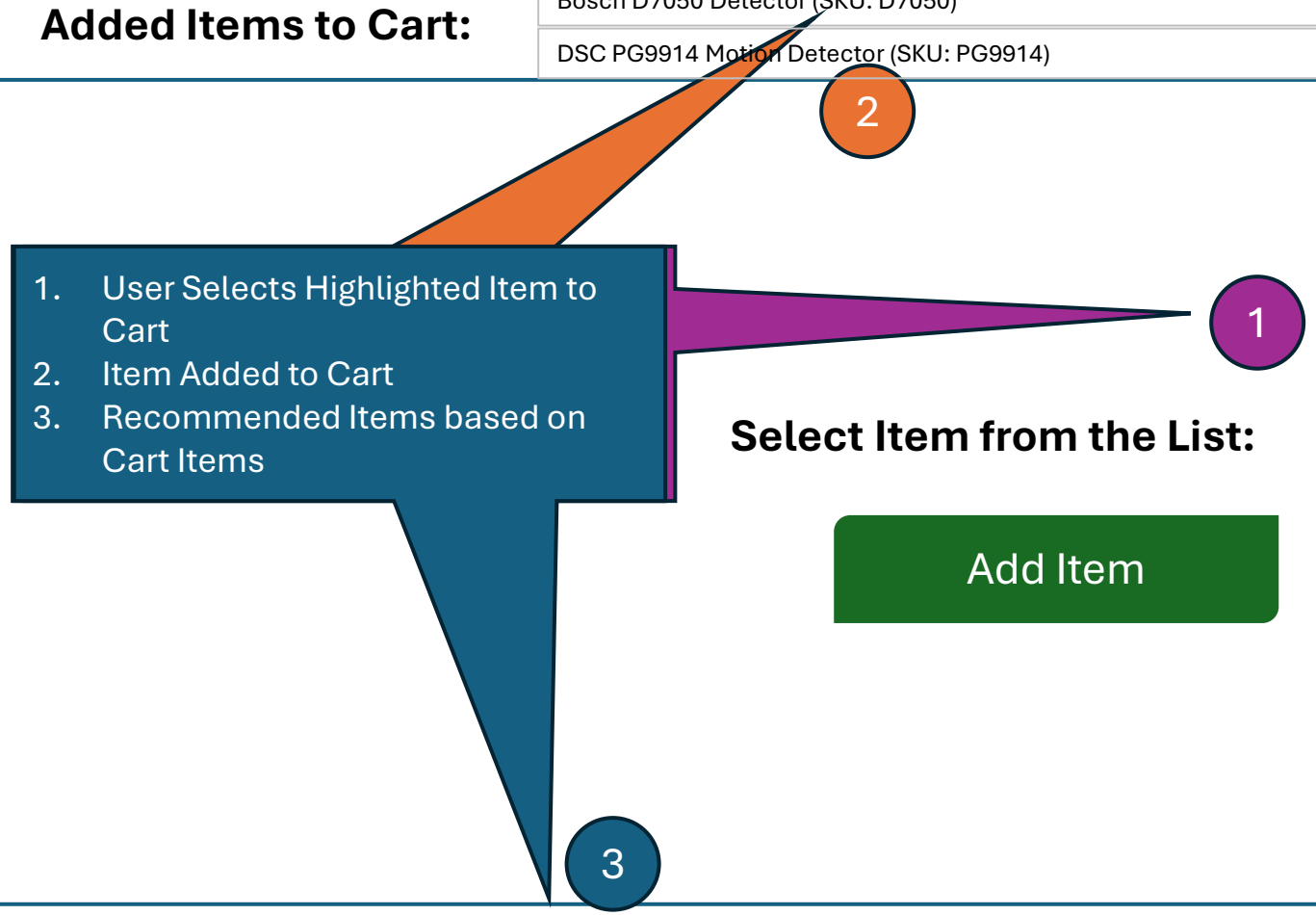
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Electronic Item Checkout Web Portal

Added Items to Cart:

Bosch D7050 Detector (SKU: D7050)

DSC PG9914 Motion Detector (SKU: PG9914)



Select Item from the List:

Add Item

Axis M3046-V Network Camera (SKU: 0806-001)
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Axis P3367-VE Network Camera (SKU: 0407-001)
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Recommended Items:

Next Items list based on Bosch D7050 Detector (SKU: D7050) and DSC PG9914 Motion Detector (SKU: PG9914)